

Chapter 4

EQUATIONS OF FIRST ORDER BUT OF HIGHER DEGREE

chapter

1. Solve: $p^3 - 6p^2 + 11p$

Solution

The given equation is

From $p - 1 = 0$, we get

From $p - 2 = 0$, we get

From $p - 3 = 0$, we get

Hence, the general solution is

2. Solve: $p(p - y) = x$

Solution

The equation can be written as

4.1 Introduction

A first order differential equation is of the form $f(x, y, p) = 0$, where $p = y'$. If the degree of p in the equation is greater than one, say n (a positive integer), then it can be expressed as

$$P_0 p^n + P_1 p^{n-1} + P_2 p^{n-2} + \dots + P_n = 0, \quad (4.1)$$

where $P_0, P_1, P_2, \dots, P_n$ are functions of x, y and $P_0 \neq 0$.

There are three types of equations are called *first order non-linear equations* and can be solved if these are of the following types:

- (i) Equations solvable for p ,
- (ii) Equations solvable for x ,
- (iii) Equations solvable for y ,
- (iv) Equations homogeneous in x, y .

4.2 Equations Solvable for p

If the equation (4.1) is solvable for p , then the expression in the left-hand side of (4.1) can be factorized into n linear factors. Thus the equation (4.1) reduces to

$$(p - f_1)(p - f_2)(p - f_3) \dots (p - f_n) = 0,$$

where $f_k = f_k(x, y), k = 1, 2, 3, \dots, n$ are functions of x, y . From the above equation we get

$$\frac{dy}{dx} = f_k(x, y), k = 1, 2, 3, \dots, n.$$

Solving we get $g_k(x, y, C) = 0, k = 1, 2, 3, \dots, n$.

Thus the general solution of the equation is

$$g_1(x, y, C) g_2(x, y, C) g_3(x, y, C) \dots g_n(x, y, C) = 0.$$

This is a linear equation. Multiplying the above

Integrating, we get

Hence the general solution is

3. Solve: $x + y = 0$

Solution

We rewrite the equation as

ILLUSTRATIVE EXAMPLES

1. Solve: $p^3 - 6p^2 + 11p - 6 = 0$.

*Solution*The given equation can be written as $(p-1)(p-2)(p-3) = 0$.From $p-1=0$, we get $dy-dx=0$. Integrating we get $y-x+C=0$.From $p-2=0$, we get $dy-2dx=0$. Integrating we get $y-2x+C=0$.From $p-3=0$, we get $dy-3dx=0$. Integrating we get $y-3x+C=0$.Hence, the general solution is $(y-x+C)(y-2x+C)(y-3x+C)=0$.

2. Solve: $p(p-y)=x(x+y)$.

Solution

The equation can be written as

$$p^2 - x^2 - py - xy = 0$$

or, $(p-x)(p+x) - y(p+x) = 0$

or, $(p-x-y)(p+x) = 0$.

From $p+x=0$, we get

$$dy + xdx = 0.$$

Integrating we get $y + \frac{x^2}{2} + C = 0$.

From $p-x-y=0$, we get

$$\frac{dy}{dx} - y = x.$$

This is a linear differential equation whose integrating factor is $e^{-\int dx} = e^{-x}$. Multiplying the above equation by e^{-x} , we get

$$\frac{d}{dx}(ye^{-x}) = xe^{-x}.$$

Integrating, we get $ye^{-x} = \int xe^{-x} dx = -(x+1)e^{-x} + C$

or, $y = -x-1 + Ce^x$.

Hence the general solution is

$$\left(y + \frac{x^2}{2} + C\right)(y + x + 1 - Ce^x) = 0.$$

3. Solve: $x + yp^2 = p(1 + xy)$.

Solution

We rewrite the equation as

$$yp^2 - p - pxy + x = 0$$

or, $(yp-1)(p-x) = 0$.

From $p - x = 0$ we get $dy = xdx$. Integrating, we get

$$y = \frac{x^2}{2} + \frac{C}{2}.$$

From $yp - 1 = 0$, we get $ydy - dx = 0$. Integrating, we get

$$x = \frac{y^2}{2} + \frac{C}{2}.$$

Hence, the general solution is

$$(2y - x^2 - C)(2x - y^2 - C) = 0.$$

4. Solve: $xy(p^2 - 1) = (x^2 - y^2)p$.

Solution

We rewrite the equation as

$$xyp^2 - x^2p + y^2p - xy = 0$$

$$\text{or, } (yp - x)(xp + y) = 0.$$

From $yp - x = 0$, we get $xdx - ydy = 0$. Integrating, we get

$$x^2 - y^2 - 2C = 0.$$

From $xp + y = 0$, we get $xdy + ydx = 0$. Integrating, we get

$$xy - C = 0.$$

Hence the general solution is

$$(x^2 - y^2 - 2C)(xy - C) = 0.$$

5. Solve: $p^2 + 2p \cot x - y^2 = 0$.

Solution

We rewrite the equation as

$$p^2 + 2p \cot x + y^2 \cot^2 x = y^2 + y^2 \cot^2 x$$

$$\text{or, } (p + y \cot x)^2 = y^2 \operatorname{cosec}^2 x$$

$$\text{or, } p = -y(\cot x \pm \operatorname{cosec} x).$$

From $p = -y(\cot x + \operatorname{cosec} x)$, we get

$$\frac{dy}{y} = -(\cot x + \operatorname{cosec} x) dx.$$

Integrating, we get

$$\ln y = -\ln \sin x - \ln(\operatorname{cosec} x - \cot x) + \ln C$$

$$\text{or, } \ln y = \ln C - \ln(1 - \cos x)$$

$$\text{or, } y(1 - \cos x) = C.$$

Similarly, from $p = -y(\cot x - \operatorname{cosec} x)$, we get

$$y(1 + \cos x) = C.$$

So the general solution is

$$[y(1 - \cos x) - C][y(1 + \cos x) - C] = 0$$

6. Solve: $3p^2y^2 - 2xyp + 4y^2 = x^2$.

Solution

We rewrite the equation as

$$9p^2y^2 - 6xyp + 4y^2 = x^2$$

$$\text{or, } (3py - x)^2 = x^2$$

$$\text{or, } p = \frac{1}{3y} [x \pm x]$$

$$\text{or, } 3ydy - xdx = 0$$

$$\text{or, } \frac{1}{2} d(x^2 - 3y^2) = 0$$

$$\text{or, } d(x^2 - 3y^2) = 0$$

Integrating, we get

$$x^2 - 3y^2 = C$$

Squaring both sides, we get

$$x^2 - 3y^2 = C$$

$$\text{or, } 3x^2 - y^2 = C$$

$$\text{or, } x^2 - 3y^2 = C$$

7. Solve: $x + \frac{p}{\sqrt{1+p^2}} = a$.

Solution

We rewrite the equation as

$$p = \frac{x - a}{\sqrt{1 - (x - a)^2}}$$

Integrating, we get

$$-(y + C) = 0$$

4.3 Equations Solved

In this case, equation

Differentiating both

$$\frac{1}{p}$$

So the general solution is

$$[y(1 - \cos x) - C][y(1 + \cos x) - C] = 0.$$

6. Solve: $3p^2y^2 - 2xyp + 4y^2 = x^2$.

Solution

We rewrite the equation as

$$9p^2y^2 - 6xyp + x^2 = 4(x^2 - 3y^2)$$

$$\text{or, } (3py - x)^2 = 4(x^2 - 3y^2)$$

$$\text{or, } p = \frac{1}{3y} \left[x \pm 2\sqrt{x^2 - 3y^2} \right]$$

$$\text{or, } 3ydy - xdx = \pm 2\sqrt{x^2 - 3y^2} dx$$

$$\text{or, } \frac{1}{2}d(x^2 - 3y^2) = \mp 2\sqrt{x^2 - 3y^2} dx$$

$$\text{or, } \frac{d(x^2 - 3y^2)}{\sqrt{x^2 - 3y^2}} = \mp 4dx.$$

Integrating, we get

$$\sqrt{x^2 - 3y^2} = \mp 2x + 3C.$$

Squaring both sides, we get

$$x^2 - 3y^2 = 4x^2 \mp 12xC + 9C^2$$

$$\text{or, } 3x^2 + 3y^2 \mp 12xC + 9C^2 = 0$$

$$\text{or, } x^2 + y^2 \mp 4xC + 3C^2 = 0.$$

7. Solve: $x + \frac{p}{\sqrt{1 + p^2}} = a$.

Solution

We rewrite the equation as

$$p = \frac{x - a}{\sqrt{1 - (x - a)^2}} \quad \text{or, } dy = \frac{x - a}{\sqrt{1 - (x - a)^2}} dx.$$

Integrating, we get

$$-(y + C) = \sqrt{1 - (x - a)^2} \quad \text{or, } (x - a)^2 + (y + C)^2 = 1.$$

4.3 Equations Solvable for x

In this case, equation (4.1) can be written as

$$x = f(y, p).$$

Differentiating both sides with respect to y, we get

$$\frac{1}{p} = \frac{dx}{dy} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial p} \frac{dp}{dy} = F\left(y, p, \frac{dp}{dy}\right).$$

This is an equation in y, p . If $y = g(p, c)$ be its solution, then using it in $x = f(y, p)$ we get

$$x = G(p, c), y = g(p, c).$$

These two give the parametric form of the general solution of the given equation. If p is eliminated from these two, we get the required solution in the form of a function in x, y .

If the equation does not contain y and is solvable for x , then it can be expressed as

$$x = f(p).$$

So the above method can be applied to solve it. On the other hand, if it can be expressed as

$$p = f(x).$$

Then the solution is $y = \int f(x) dx + C$.

ILLUSTRATIVE EXAMPLES

1. Solve: $x = y + ap^2$.

Solution

Differentiating both sides with respect to y , we get

$$\frac{1}{p} = \frac{dx}{dy} = 1 + 2ap \frac{dp}{dy}$$

$$\text{or, } dy = \frac{2ap^2}{1-p} dp$$

$$\text{or, } dy = -2a(p+1)dp + 2a \frac{dp}{1-p}.$$

Integrating, we get $y = -ap^2 - 2ap - 2a \ln(p-1) + C$.

Using this in the given equation, we get $x = -2ap - 2a \ln(p-1) + C$.

The p eliminant of these two will give the general solution.

2. Solve: $x = py - ap^2$.

Solution

Differentiating both sides with respect to y , we get

$$\frac{1}{p} = \frac{dx}{dy} = p + (y - 2ap) \frac{dp}{dy}$$

$$\text{or, } \frac{dy}{dp} = \frac{yp - 2ap^2}{1 - p^2}$$

$$\text{or, } \frac{dy}{dp} + \frac{yp}{p^2 - 1} = \frac{2ap^2}{p^2 - 1}.$$

This is a linear differential equation

$$\int \frac{p dp}{p^2 - 1} =$$

Multiplying the above equation by

$$\frac{d}{dp} \{ y \sqrt{p^2 - 1} \}$$

Integrating, we get

$$y \sqrt{p^2 - 1} = a$$

$$\text{or, } y = ap + \frac{(C + a)}{\sqrt{p^2 - 1}}$$

Using this in the given equation

$$x = \frac{p(C + a)}{\sqrt{p^2 - 1}}$$

The p eliminant of these two

3. Solve: $y - 2px + ayp^2 = 0$.

Solution

We rewrite the equation as

Differentiating both sides

or,

or,

or,

Integrating, we get

Eliminating p from the

or,

This is a linear differential equation, whose integrating factor is

$$e^{\int \frac{p dp}{p^2-1}} = e^{\frac{1}{2} \ln(p^2-1)} = \sqrt{p^2-1}.$$

Multiplying the above equation by $\sqrt{p^2-1}$, we get

$$\frac{d}{dp} \left\{ y \sqrt{p^2-1} \right\} = \frac{2ap^2}{\sqrt{p^2-1}} = 2a \left\{ \sqrt{p^2-1} + \frac{1}{\sqrt{p^2-1}} \right\}.$$

Integrating, we get

$$y \sqrt{p^2-1} = a \left\{ p \sqrt{p^2-1} + \cosh^{-1} p \right\} + C$$

$$\text{or, } y = ap + \frac{(C + a \cosh^{-1} p)}{\sqrt{p^2-1}}.$$

Using this in the given equation, we get

$$x = \frac{p(C + a \cosh^{-1} p)}{\sqrt{p^2-1}}$$

The p eliminant of these two will give the general solution.

3. Solve: $y - 2px + ayp^2 = 0$.

Solution

We rewrite the equation as

$$x = \frac{y}{2p} + \frac{ayp}{2}.$$

Differentiating both sides with respect y , we get

$$\frac{1}{p} = \frac{dx}{dy} = \frac{1}{2p} - \frac{y}{2p^2} \frac{dp}{dy} + \frac{ap}{2} + \frac{ay}{2} \frac{dp}{dy}$$

$$\text{or, } \frac{dp}{dy} \left(\frac{ay}{2} - \frac{y}{2p^2} \right) = \frac{1}{2p} - \frac{ap}{2}$$

$$\text{or, } \frac{dp}{dy} \frac{y(ap^2-1)}{2p^2} = \frac{1-ap^2}{2p}$$

$$\text{or, } \frac{dp}{p} + \frac{dy}{y} = 0. \text{ if } 1-ap^2 \neq 0.$$

Integrating, we get $py = C$.

Eliminating p from the given equation with the help of above equation, we get

$$y - \frac{2xC}{y} + ay \frac{C^2}{y^2} = 0$$

$$\text{or, } y^2 = 2xC - aC^2.$$

If $ap^2 = 1$, then from the given equation we get

$$y = px \text{ or, } x = \sqrt{ay}.$$

This solution of the given equation neither contains any arbitrary constant nor can be derived from the general solution, given above, by assigning any particular value to the arbitrary constant. This is called singular solution of the equation.

4. Solve: $p^2 - 2xp + 1 = 0$.

Solution

We rewrite the equation as

$$x = \frac{1}{2} \left(p + \frac{1}{p} \right).$$

Differentiating with respect to y , we get

$$\frac{1}{p} = \frac{dx}{dy} = \frac{1}{2} \left(1 - \frac{1}{p^2} \right) \frac{dp}{dy}$$

or, $dy = \frac{1}{2} \left(p - \frac{1}{p} \right) dp.$

Integrating, we get

$$y = \frac{1}{4} p^2 - \frac{1}{2} \ln p + C.$$

Hence, p -eliminant of $x = \frac{1}{2} \left(p + \frac{1}{p} \right)$ and $y = \frac{1}{4} p^2 - \frac{1}{2} \ln p + C$ give the general solution.

5. Solve: $p^3 - 4xyp + 8y^2 = 0$.

Solution

We rewrite the equation as

$$x = \frac{p^2}{4y} + \frac{2y}{p}.$$

Differentiating with respect to y , we get

$$\frac{1}{p} = \frac{dx}{dy} = -\frac{1}{4} \frac{p^2}{y^2} + \left(\frac{p}{2y} - \frac{2y}{p^2} \right) \frac{dp}{dy} + \frac{2}{p}$$

or, $\frac{1}{4} \frac{p^2}{y^2} - \frac{1}{p} = \left(\frac{p}{2y} - \frac{2y}{p^2} \right) \frac{dp}{dy}$

or, $\frac{dy}{2y} = \frac{dp}{p}$, if, $p^3 \neq 4y^2$.

Integrating, we get

$$p^2 = 4Cy.$$

Using this in the given equation, we get

$$p(C - x) = -2y$$

or, $4Cy(C - x)^2 = 4y^2$

or, $y = C(x - C)^2.$

If $p^3 = 4y^2$, then from the given equation

$$3y = px \text{ or, } 27y$$

This solution of the given equation neither be derived from the general solution, given above, by assigning any particular value to the arbitrary constant. This is called singular solution.

6. Solve: $x = 4p + 4p^3$.

Solution

Differentiating with respect to y , we get

$$\frac{1}{p} = \frac{dx}{dy} = 4 \left(1 + 3p^2 \right)$$

or, $dy = 4(p + 3p^3)$

Integrating, we get $y = 2p^2 + 3p^4$

Hence, p -eliminant of $x = 4p + 4p^3$ solution

4.4 Equations Solvable for y

In this case, equation (1) can be written as

$$y = f(x, p)$$

Differentiating both sides with respect to x , we get

$$p = \frac{dy}{dx} =$$

This is an equation in x, p . If $x = g(p, c)$, we get $y = G(p, c), x = g(p, c)$.

These two give the parametric form of the solution. If p is eliminated from these two, we get the solution in x, y .

Sometimes it is possible to express the solution as $f(x, p) = px + g(p)$.

A differential equation of the form

Differentiating this with respect to x , we get

$$p = \frac{dy}{dx} = p + x \frac{dp}{dx} + g'(p)$$

This gives either $\frac{dp}{dx} = 0$ or, $x = -g'(p)$. Integrating p we see that the general solution is

If $p^3 = 4y^2$, then from the given equation we get

$$3y = px \text{ or, } 27y = 4x^3.$$

This solution of the given equation neither contains any arbitrary constant nor can be derived from the general solution, given above, by assigning any particular value to the arbitrary constant. This is called singular solution of the equation.

6. Solve: $x = 4p + 4p^3$.

Solution

Differentiating with respect to y , we get

$$\frac{1}{p} = \frac{dx}{dy} = 4(1 + 3p^2) \frac{dp}{dy}$$

$$\text{or, } dy = 4(p + 3p^3) dp.$$

Integrating, we get $y = 2p^2 + 3p^4 + C$.

Hence, p -eliminant of $x = 4p + 4p^3$ and $y = 2p^2 + 3p^4 + C$ will give the general solution

4.4 Equations Solvable for y

In this case, equation (1) can be written as

$$y = f(x, p).$$

Differentiating both sides with respect to x , we get

$$p = \frac{dy}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial p} \frac{dp}{dx} = F\left(x, p, \frac{dp}{dx}\right).$$

This is an equation in x, p . If $x = g(p, c)$ be its solution, then using it in $y = f(x, p)$, we get $y = G(p, c), x = g(p, c)$.

These two give the parametric form of the general solution of the given equation. If p is eliminated from these two, we get the required solution in the form of a function in x, y .

Sometimes it is possible to express the function $f(x, p)$ in the equation $y = f(x, p)$, as $f(x, p) = px + g(p)$,

A differential equation of the form $y = px + g(p)$, is known as Clairaut's equation.

Differentiating this with respect to x , we get

$$p = \frac{dy}{dx} = p + x \frac{dp}{dx} + g'(p) \frac{dp}{dx} \text{ or, } \{x + g'(p)\} \frac{dp}{dx} = 0,$$

This gives either $\frac{dp}{dx} = 0$ or, $x + g'(p) = 0$. From $\frac{dp}{dx} = 0$, we get $p = C$ and eliminating p we see that the general solution of the differential equation of Clairaut's form is

$$y = Cx + g(C).$$

When $x + g'(p) = 0$, then eliminating from the p , with the aid of it, from the given equation we get a relation between x, y without containing any constant and this relation can not be obtained from the general solution by giving a particular value to the arbitrary constant. Such a solution is known as *singular solution*. It can be easily shown that these solutions are linearly independent.

It is to be mentioned that a differential equation, which is not given in Clairaut's form, can be reduced to Clairaut's equation under some suitable substitution. So the above method can be applied to solve the reduced equation and after solving it and using inverse substitution, we get the solution of the original equation.

Further, if the function $f(x, p)$ in the equation $y = f(x, p)$, has the following form

$$f(x, p) = x\phi(p) + g(p),$$

then the differential equation takes the form

$$y = x\phi(p) + g(p),$$

which is known as *Lagrange's equation*.

If $\phi(p) \neq p$, then differentiating with respect to x , we get

$$p = \frac{dy}{dx} = \phi(p) + \{x\phi'(p) + g'(p)\} \frac{dp}{dx}$$

$$\text{or, } \frac{dx}{dp} = \frac{x\phi'(p) + g'(p)}{p - \phi(p)}$$

$$\text{or, } \frac{dx}{dp} - \frac{\phi'(p)}{p - \phi(p)} x = \frac{g'(p)}{p - \phi(p)},$$

which is a linear differential equation and can be solved easily. The p eliminant of the solution of this equation and the given equation give the general solution.

If $\phi(p) = p$, then the above equation reduces to Clairaut's form. Thus *Lagrange's equation* is an extended form of the Clairaut's equation.

If the equation does not contain x and is solvable for y , then it can be expressed as

$$y = f(p).$$

So the above method can be applied to solve it. On the other hand, if it can be expressed as

$$p = f(y),$$

$$\text{i.e., } \frac{dy}{dx} = f(y),$$

then the solution is given by

$$\int \frac{dy}{f(y)} = x + C.$$

1. Solve: $y = p^2x + p$.

Solution

Differentiating the given equation

$$p = \frac{dy}{dx} =$$

$$\text{or, } p(1 - p) \frac{dp}{dx} =$$

$$\text{or, } \frac{dx}{dp} + \frac{2x}{p-1} =$$

This is a linear differential equation

$$e^{\int \frac{2dp}{p-1}} =$$

Multiplying the above equation

$$\frac{d}{dp} [x(p-1)^2] =$$

Integrating, we get $x(p-1)^2 =$

$$\text{or, } x = \frac{1}{(p-1)^2}$$

Hence p -eliminant of $x = \frac{1}{(p-1)^2}$

2. Solve: $y = 2px + p^2$

Solution

Differentiating both sides

$$p = \frac{dy}{dx} = 2p +$$

This is a linear differential equation. Multiplying the above equation

Integrating, we get

$$xp^2 =$$

ILLUSTRATIVE EXAMPLES

1. Solve: $y = p^2x + p$.

Solution

Differentiating the given equation with respect to x , we get

$$p = \frac{dy}{dx} = p^2 + (2px + 1) \frac{dp}{dx}$$

$$\text{or, } p(1 - p) \frac{dx}{dp} = (2px + 1)$$

$$\text{or, } \frac{dx}{dp} + \frac{2x}{p-1} = \frac{1}{p(1-p)}$$

This is a linear differential equation whose integrating factor is

$$e^{\int \frac{2dp}{p-1}} = e^{2 \ln(p-1)} = (p-1)^2.$$

Multiplying the above equation by $(p-1)^2$, we get

$$\frac{d}{dp} [x(p-1)^2] = \frac{1}{p} - 1.$$

Integrating, we get $x(p-1)^2 = \ln p - p + C$

$$\text{or, } x = \frac{\ln p - p + C}{(p-1)^2}.$$

Hence p -eliminant of $x = \frac{\ln p - p + C}{(p-1)^2}$ and $y = p^2x + p$ will give the solution.

2. Solve: $y = 2px + p^2$

Solution

Differentiating both sides with respect to x , we get

$$p = \frac{dy}{dx} = 2p + 2(x + p) \frac{dp}{dx} \quad \text{or, } \frac{dx}{dp} + \frac{2x}{p} = -2.$$

This is a linear differential equation whose integrating factor is $e^{\int \frac{2}{p} dp} = p^2$. Multiplying the above equation by p^2 , we get

$$\frac{d}{dp} (xp^2) = -2p^2.$$

Integrating, we get

$$xp^2 = -\frac{2}{3} p^3 + C \quad \text{or, } x = -\frac{2}{3} p + Cp^{-2}.$$

Using this in the given equation, we get

$$y = -\frac{1}{3}p^2 + \frac{2C}{p}$$

Hence, the p -eliminant of $x = -\frac{2}{3}p + Cp^{-2}$ and $y = -\frac{1}{3}p^2 + \frac{2C}{p}$ will give the general solution.

3. Solve: $y = \ln(p^3 + p)$.

Solution

Differentiating with respect to x , we get

$$p = \frac{dy}{dx} = \frac{3p^2 + 1}{p^3 + p} \frac{dp}{dx} \quad \text{or, } dx = \frac{3p^2 + 1}{(p^2 + 1)p^2} dp$$

Integrating, we get

$$x = 2 \tan^{-1} p - \frac{1}{p} + C.$$

Hence, the p -eliminant of $x = 2 \tan^{-1} p - \frac{1}{p} + C$ and $y = \ln(p^3 + p)$ will give the general solution.

4. Solve: $y = px + \sqrt{a^2 p^2 + b^2}$.

Solution

The given equation is in Clairaut's form. Differentiating both sides with respect to x , we get

$$p = \frac{dy}{dx} = p + \left(x + \frac{a^2 p}{\sqrt{a^2 p^2 + b^2}} \right) \frac{dp}{dx}$$

$$\text{or, } \left(x + \frac{a^2 p}{\sqrt{a^2 p^2 + b^2}} \right) \frac{dp}{dx} = 0.$$

From $\frac{dp}{dx} = 0$, we get $p = C$. Using this in the given equation we get general solution as

$$y = Cx + \sqrt{a^2 C^2 + b^2}.$$

From $x + \frac{a^2 p}{\sqrt{a^2 p^2 + b^2}} = 0$, we get $x = -\frac{a^2 p}{\sqrt{a^2 p^2 + b^2}}$. Eliminating p from the given equation with the aid of it we get the singular solution as

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{a^2 p^2}{a^2 p^2 + b^2} + \left(-\frac{a^2 p^2}{\sqrt{a^2 p^2 + b^2}} + \sqrt{a^2 p^2 + b^2} \right)^2 \frac{1}{b^2} = 1.$$

5. Reduce the differential equation $xy(y - px)$ substitution $x^2 = u$, $y^2 = v$ and then solve it.

Solution

From $x^2 = u$, $y^2 = v$, we get $\frac{ydy}{xdx} = \frac{dv}{du}$. W

$$xv - uxP = x + xP \quad \text{or, } v =$$

This is Clairaut's form. So the general sol

$$v = uC + C + 1 \quad \text{or, } y^2 =$$

6. Reduce the differential equation $(y + xp)$ substitution $xy = v$ and then solve it.

Solution

From $xy = v$, we get $xp + y = \frac{dv}{dx} = P(s$

$$P^2 = x(P - y) = xP$$

This is in Clairaut's form, So the gene

$$v = xC - C^2 \quad \text{or, } y$$

7. Reduce the different equation $y^2 \ln y$ substitution $\ln y = v$ and then solve it.

Solution

From $\ln y = v$, we get $\frac{p}{y} = \frac{dv}{dx} = P$

$$y^2 v = xy^2 P + y$$

This is in Clairaut's form. So the g

$$v = xC + C^2 \quad \text{or, } y$$

8. Reduce the differential equation

$$(x^2 + y^2)(1 + p)^2 - 2(x + y)(1 + p)$$

the substitution $x + y = u$, $x^2 + y^2 =$

Solution

Form $x + y = u$, $x^2 + y^2 = v$ we

$$2xdx + 2ydy =$$

Dividing we get

$$P = \frac{d}{d}$$

5. Reduce the differential equation $xy(y - px) = x + py$ to Clairaut's form by the substitution $x^2 = u$, $y^2 = v$ and then solve it.

Solution

From $x^2 = u$, $y^2 = v$, we get $\frac{ydy}{xdx} = \frac{dv}{du}$. Writing $\frac{dy}{dx} = p$ and $\frac{dv}{du} = P$, we get

$$xv - uxP = x + xP \text{ or, } v = uP + P + 1.$$

This is Clairaut's form. So the general solution is

$$v = uC + C + 1 \text{ or, } y^2 - 1 = C(x^2 + 1).$$

6. Reduce the differential equation $(y + xp)^2 = x^2p$ to Clairaut's form by the substitution $xy = v$ and then solve it.

Solution

From $xy = v$, we get $xp + y = \frac{dv}{dx} = P$ (say). Using this in the given equation, we get

$$P^2 = x(P - y) = xP - v \text{ or, } v = xP - P^2.$$

This is in Clairaut's form, So the general solution is

$$v = xC - C^2 \text{ or, } xy = xC - C^2.$$

7. Reduce the differential equation $y^2 \ln y = xyp + p^2$ to Clairaut's form by the substitution $\ln y = v$ and then solve it.

Solution

From $\ln y = v$, we get $\frac{p}{y} = \frac{dv}{dx} = P$ (say). Using this in the given equation we get

$$y^2v = xy^2P + y^2P^2 \text{ or, } v = xP + P^2.$$

This is in Clairaut's form. So the general solution is

$$v = xC + C^2 \text{ or, } \ln y = xC + C^2.$$

8. Reduce the differential equation

$$(x^2 + y^2)(1 + p)^2 - 2(x + y)(1 + p)(x + yp) + (x + yp)^2 = 0 \text{ to Clairaut's form by}$$

the substitution $x + y = u$, $x^2 + y^2 = v$ and then solve it.

Solution

Form $x + y = u$, $x^2 + y^2 = v$ we get

$$2xdx + 2ydy = dv \text{ and } dx + dy = du.$$

Dividing we get

$$P = \frac{dv}{du} = \frac{2(x + yp)}{1 + p}.$$